Examiner only

2. A car is travelling at 10 m/s along a flat road. A driving force of 50 000 N acts on the car for a distance of 20 m causing it to accelerate.

(a) (i) Use an equation from page 2 to calculate the work done by the 50 000 N force. [2]

Work done = ... J

(ii) The car gains 600 000 J of kinetic energy as it accelerates. Calculate how much energy is transferred in other ways. [1]

Energy transfer = ... J

(b) State **two** ways that the **design** of the car could be changed to improve its efficiency. [2]

1. ....

2. ....

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